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Evaluating compressive strength of concrete made with recycled concrete aggregates using machine learning approach

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ABSTRACT

To reduce the environmental impact of construction and demolition waste of concrete, recycled concrete aggregate (RCA) has been widely utilized in concrete. The compressive strength of recycled concrete is one of the most important parameters governing the quality of concrete. The compressive strength is determined from the compression test, which requires a huge amount of materials as well as consumes cost and time. Thus, to solve those limitations, this study focused on evaluating the compressive strength of concrete made from RCA using different single and hybrid models of machine learning. Six machine learning models including Gradient Boosting (GB), Extreme Gradient Boosting (XGB), Support Vector Regression (SVR), and three hybrid models of those single models with Particle Swarm Optimization (PSO) namely GB_PSO, XGB_PSO, and SVR_PSO were used to estimate the compressive strength of recycled concrete. The input variables for modeling consisted of cement content, water content, aggregate content, natural aggregate content, recycle concrete aggregate content, sand content, water absorption rates of natural aggregate and RCA. The results of this study show that hybrid models performed better than single models in terms of prediction accuracy. The results indicated that the GB_PSO has the highest prediction accuracy with R = 0.9356, RMSE = 5.5604 MPa, and MAE = 4.2882 MPa. The results of feature importance analysis and partial dependence plots (PDP) analysis revealed that the most important variable effect on compressive strength of concrete made with RAC is cement content, whatever performance strategies of concrete made with RAC. From the results of PDP, the quantity of each material can be computed easily for the designed compressive strength. In the end, this study provides a systematic evaluation of the compressive strength prediction of recycled concrete and has a significant contribution to literature and practice.

1. Introduction

Concrete is one of the most common materials used in the construction industry. Due to the rapid development of the economy as well as urbanization, the construction of new structures and the demolition of old structures have increased significantly. In Vietnam, the annual amount of waste from construction and demolition increased from 1.0 million tons in 2004 [1] to 1.9 million tons by 2011 [2]. If this waste material can be not treated properly, a huge amount of solid waste materials will cause many environmental problems. Besides, due to the population explosion, there are limited landfill areas in Vietnam, therefore, it is necessary to seek a new solution to utilize this waste material. In parallel, the construction of new projects using natural coarse and fine aggregates leads to a depletion of resources. The usage and exploitation of natural aggregate produce a negative impact on the environment. Thus, it is very crucial for the concrete industry to explore a new approach to use this waste material as an aggregate for sustainable development.

Many previous studies have successfully used recycled concrete aggregate (RCA) in the production of concrete and mortar [3-9]. In general, the replacement of natural coarse aggregate by RCA will lead to a reduction of mechanical properties of concrete such as compressive strength and tensile strength [6,9]. However, it was reported that addition to 30% RCA to concrete has a small significant impact on the strength, the high-strength concrete made with RCA has a similar performance in terms of engineering properties as well as durability

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Table 1
Database collection.

No.	Reference	Number of data	Proportion of data (%)	No.	Reference	Number of data	Proportion of data (%)
1	Ismail and Ramili [27]	40	5.55	35	Zega and Di Maio [28]	8	1.11
2	Kou et al. [29]	36	4.99	36	Yang et al. [30]	7	0.97
3	Thomas et al. [31]	36	4.99	37	Evangelista and De Brito [32]	6	0.83
4	Limbachiya et al. [7]	28	3.88	38	Fathifazl et al. [33]	6	0.83
5	Ulloa et al. [34]	25	3.47	39	Mohammed et al. [35]	6	0.83
6	Thomas et al. [36]	24	3.33	40	Somna et al. [37]	6	0.83
7	Thomas et al. [38]	24	3.33	41	Xiao et al. [39]	6	0.83
8	Thomas et al. [40]	24	3.33	42	Gomez-Sobreron [41]	5	0.69
9	Vivian et al. [42]	24	3.33	43	Manzi et al. [43]	5	0.69
10	Chaocan Zheng et al. [44]	20	2.77	44	Nepomuceno et al. [45]	5	0.69
11	Haitao and Shizhu [46]	20	2.77	45	Woubishet [47]	5	0.69
12	Sheen et al. [48]	20	2.77	46	Younis and Pilakoutas [49]	5	0.69
13	Pedro et al. [50]	18	2.50	47	Arezoumandi et al. [51]	4	0.55
14	Poon et al. [52]	17	2.36	48	Cakir and Sofyanli [53]	4	0.55
15	Butler et al. [54]	16	2.22	49	Choi et al. [55]	4	0.55
16	Duan and Poon [56]	16	2.22	50	Domingo-Cabo et al. [57]	4	0.55
17	Knaack and Kurama [58]	16	2.22	51	Etxeberria et al. [59]	4	0.55
18	Pereira et al. [60]	15	2.08	52	Etxeberria et al. [61]	4	0.55
19	SC Kou et al. [62]	15	2.08	53	Folino and Xargay [63]	4	0.55
20	Gayarre et al. [64]	14	1.94	54	Fonseca et al. [65]	4	0.55
21	Barbudo et al. [66]	12	1.66	55	Kang et al. [67]	4	0.55
22	Kim et al. [68]	11	1.53	56	Luis Ferreira et al. [69]	4	0.55
23	Andreu and Miren [70]	10	1.39	57	Rafi [71]	4	0.55
24	Beltran et al. [72]	10	1.39	58	Rao et al. [73]	4	0.55
25	Corinaldesi [74]	10	1.39	59	Sato et al. [75]	4	0.55
26	Matias et al. [76]	10	1.39	60	Wardeh et al. [77]	4	0.55
27	Rahal [78]	10	1.39	61	Cui et al. [79]	3	0.42
28	Casuccio et al. [80]	9	1.25	62	Pepe et al. [81]	3	0.42
29	Hoffmann et al. [82]	9	1.25	63	Carneiro et al. [83]	2	0.28
30	Belen et al. [84]	8	1.11	64	Dibas et al. [85]	2	0.28
31	Beltran et al. [86]	8	1.11	65	Ignjatovic et al. [87]	2	0.28
32	Gonzalez-Fonteboa et al. [88]	8	1.11	66	Kou and Poon [89]	2	0.28
33	Huda and Alam [90]	8	1.11	67	Radonjanin et al. [91]	2	0.28
34	Medina et al. [92]	8	1.11		**Total**	**721**	**100.00**

compared to the concrete using natural aggregate [7]. Furthermore, other studies indicated that concrete with RCA has comparable fresh and hardened properties to normal aggregate concrete, the ratio between splitting tensile and compressive strength was revealed to be in good alignment with that of normal aggregate [9]. In general, the compressive strength is determined in the laboratory via the compression test for the specimens at specified curing age. However, to conduct this test, it is also required a quite huge amount of material as well as cost and time. Thus, some researchers have used several empirical equations as well as a method to estimate the compressive strength of concrete using natural aggregate as well as RAC.

The compressive strength of recycled concrete has a strong relation with aggregate (natural and recycled aggregates) content, water/cement ratio, cement content, type of aggregate, and type and dosage of admixture [10,11]. Nevertheless, those parameters and compressive strength have a complicated and non-linear relationship, and there is no universal equation, which can reflect and clarify accurately their correlations [12,13]. Thus, several authors used machine learning (ML) and artificial intelligence (AI) to estimate the compressive strength of concrete using recycled aggregate [3,4,14-17]. A previous study used 168 datasets from the literature to predict the compressive strength of RCA concrete and they indicated that an artificial neural network (ANN) is a promising algorithm to estimate the compressive strength of recycled concrete considering different kinds and sources of recycled aggregates [4]. Deshpande et al. [3] also used ANN to forest cast the compressive strength of concrete containing RCA, they also revealed that this algorithm is a promising approach to estimate the compressive strength.

It was stated that the result modeled by ANN and fuzzy logic (FL) models has a close value to that of experimental results [18]. Previous studies stated that the ANN model performed better than the model tree and linear regression analysis in predicting the compressive strength of concrete with RAC [3,19]. Furthermore, Deng et al, [14] used convolutional neural networks (CNN) to estimate the compressive strength of concrete made with RCA, they revealed that the model used CNN performed better than those using conventional neural networks. Nunez et al. [20] used the Gaussian process, deep learning, and gradient boosting (GB) to estimate the compressive strength of recycled concrete, they revealed that GB produced the highest prediction accuracy with $R = 0.959$ and $MAE = 5.076$. In almost the above-mentioned published studies, they only used accuracy metrics to estimate the prediction accuracy of the model. However, only accuracy metrics may not be enough to assess the prediction accuracy of the models, and another indicator such as feature importance is also very crucial to evaluate properly the prediction accuracy of the model. Furthermore, almost previous studies used single machine learning approaches to estimate the compressive strength of concrete using RCA. However, these single machine learning methods have disadvantages in terms of evolutionary algorithms. Particle swarm optimization (PSO) is known as a powerful technique, which has been used widely for optimization problems in civil engineering problems [21-23]. The PSO has been recognized as an optimization technique, which has no complicated operators as evolutionary algorithms because it has fewer parameters that need to be adjusted [24]. Thus, the PSO has been used to improve and optimize the prediction ability of the hybrid model as reported in previous studies [25,26].

Therefore, this study used six different machine learning methods to estimate the compressive strength of concrete using RCA as a partial coarse aggregate replacement. The six machine learning models including single models (GB, extreme gradient boosting (XGB), and support vector machine regression (SVR)) and hybrid models of GB_PSO, XGB_PSO, and SVR_PSO, were employed for modeling. The input parameter used in this study comprises cement content, water content, aggregate content (including natural and RCA), sand content, water absorption of natural aggregate and RCA. This state-of-the-art research contributes valuable knowledge to literature and practive as follows.

Table 2
Description of database used in this study.

	Unit	count	Min	$Q_{25\%}$	Average	Median	$Q_{75\%}$	Max	Std	Skw
Water content	kg/m^3	721	76.00	162.25	181.35	184.00	197.60	245.00	27.97	−0.30
Aggregate content	kg/m^3	721	530.39	962.00	1065.73	1039.50	1120.00	1950.00	196.52	1.56
RAC content	kg/m^3	721	0.00	187.00	531.56	495.00	920.70	1836.10	426.16	0.47
NA content	kg/m^3	721	0.00	0.00	534.17	540.00	890.40	1950.00	435.65	0.12
Sand content	kg/m^3	721	400.00	650.00	725.07	701.75	800.00	1283.33	132.71	0.43
Cement content	kg/m^3	721	210.00	333.00	380.68	380.00	410.00	650.00	76.84	0.79
RCAWA	%	721	1.24	3.90	5.42	5.30	6.40	14.90	2.11	1.47
NAWA	%	721	0.10	0.70	1.09	1.00	1.30	3.00	0.58	1.04
Compressive strength	MPa	721	13.40	34.70	44.15	43.00	51.50	108.51	14.39	0.94

Sk = Skewness; Std = Standard deviation;

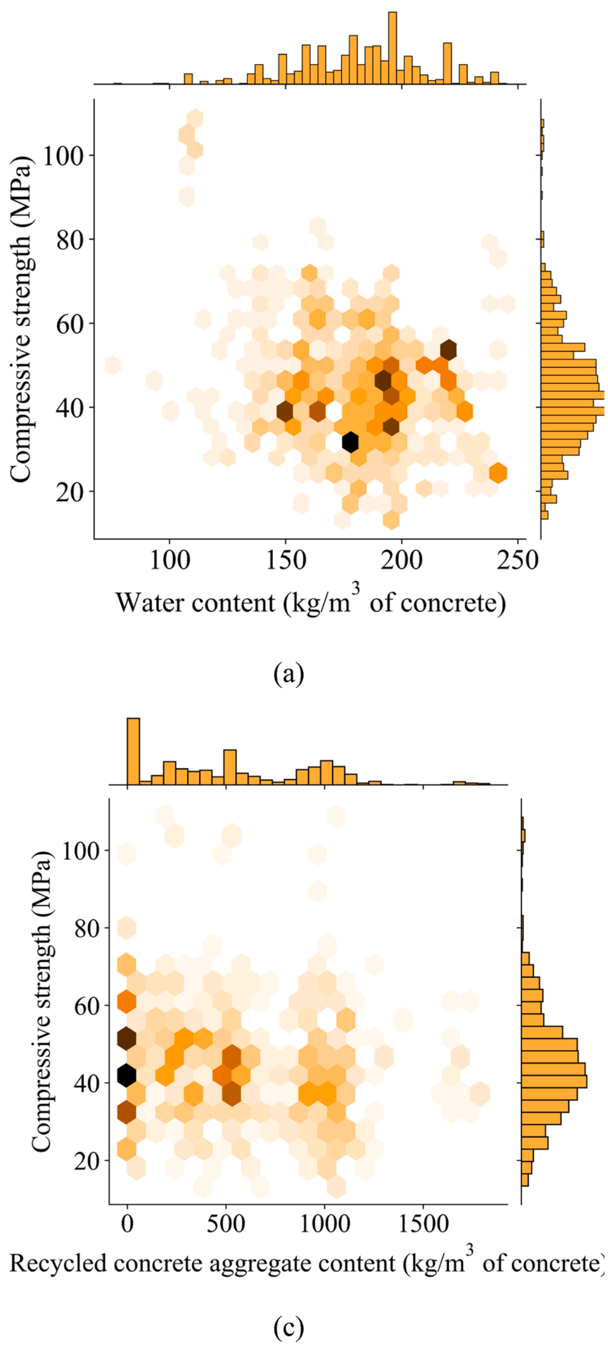


Fig. 1. Hex contour graph of input parameters (a) water content, (b) aggregate content, (c) RCA content, (d) NA content, (e) sand content, (f) cement content, (g) RCAWA, and (h) NAWA.

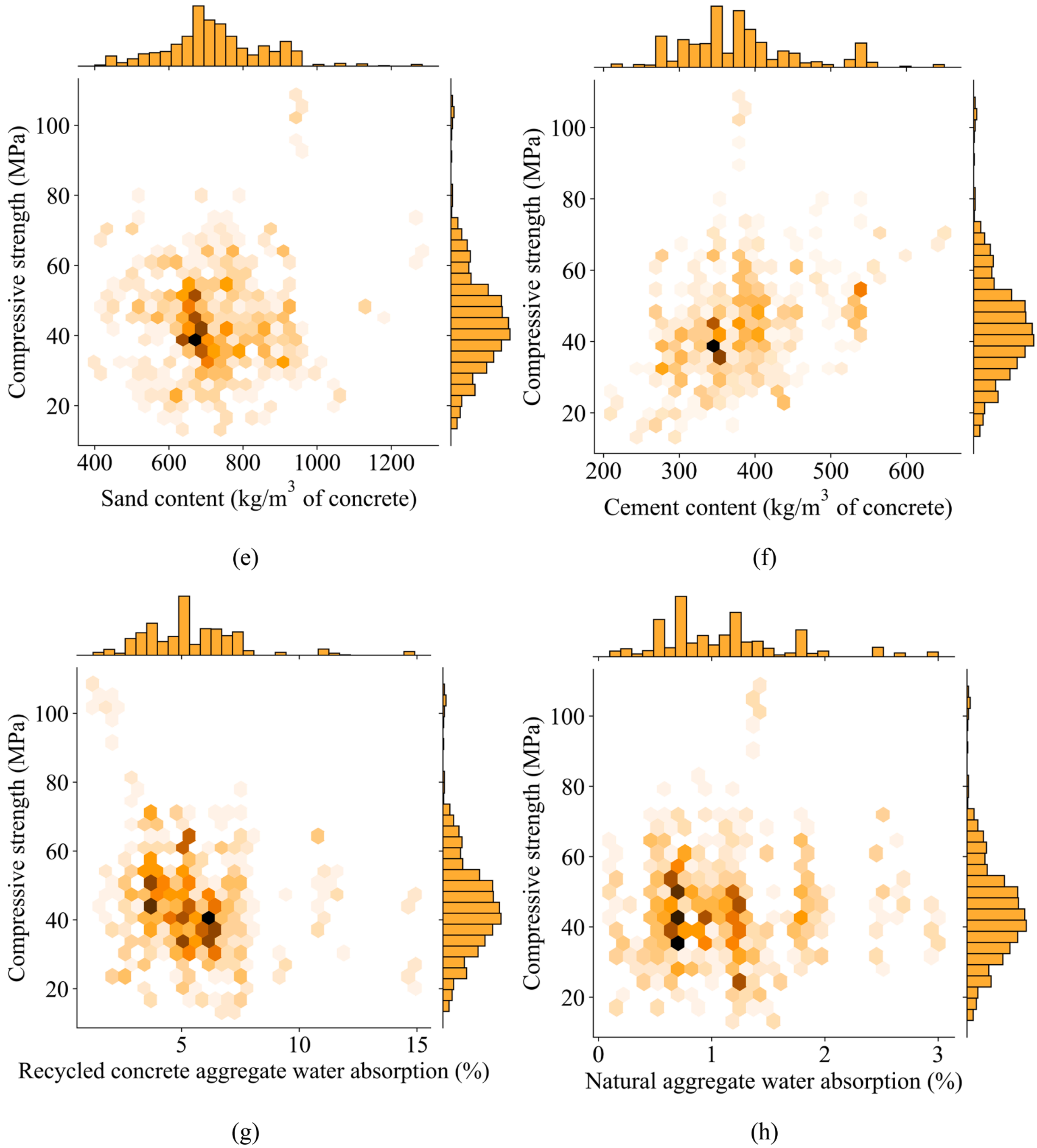


Fig. 1. (*continued*).

First, this study can fill the gap in the literature related to the application of ML model (particular hybrid-model) in evaluating the compressive strength of concrete using RCA. Second, this study used both using Shapley Additive exPlanations (SHAP) and Partial dependence plots (PDP 1D and 2D) to assess the importance and the influence of each and coupled variables on the prediction of the compressive strength. Third, based on the results of PDP analysis, the practical engineering application related to design mix proportion for predicting compressive strength was proposed. Fourth, statistical indicators were also employed to completely evaluate the performance of the models for both training, testing, and all datasets. Finally, this study will fill the gap of literate in estimating the compressive strength of concrete using RCA using the machine learning approach. Thus, it can be indicated that this is the first and novel research to systematically predict the compressive strength of recycled concrete.

2. Database description and analysis

In this study, a total of 721 data samples of experimental results collected from 67 literature was used to train and test the machine learning model for predicting the compressive strength of concrete using recycled aggregate. Table 1 shows the detail of source references and the number of data in each reference as well as the percentage of data in each reference contributing to the total data. Table 2 presents the database used in this study, a total of 8 input variables were employed to build the machine learning models. The input variables consist of the main input parameters, which strongly affect the compressive strength of concrete. These input variables are water content, cement content,

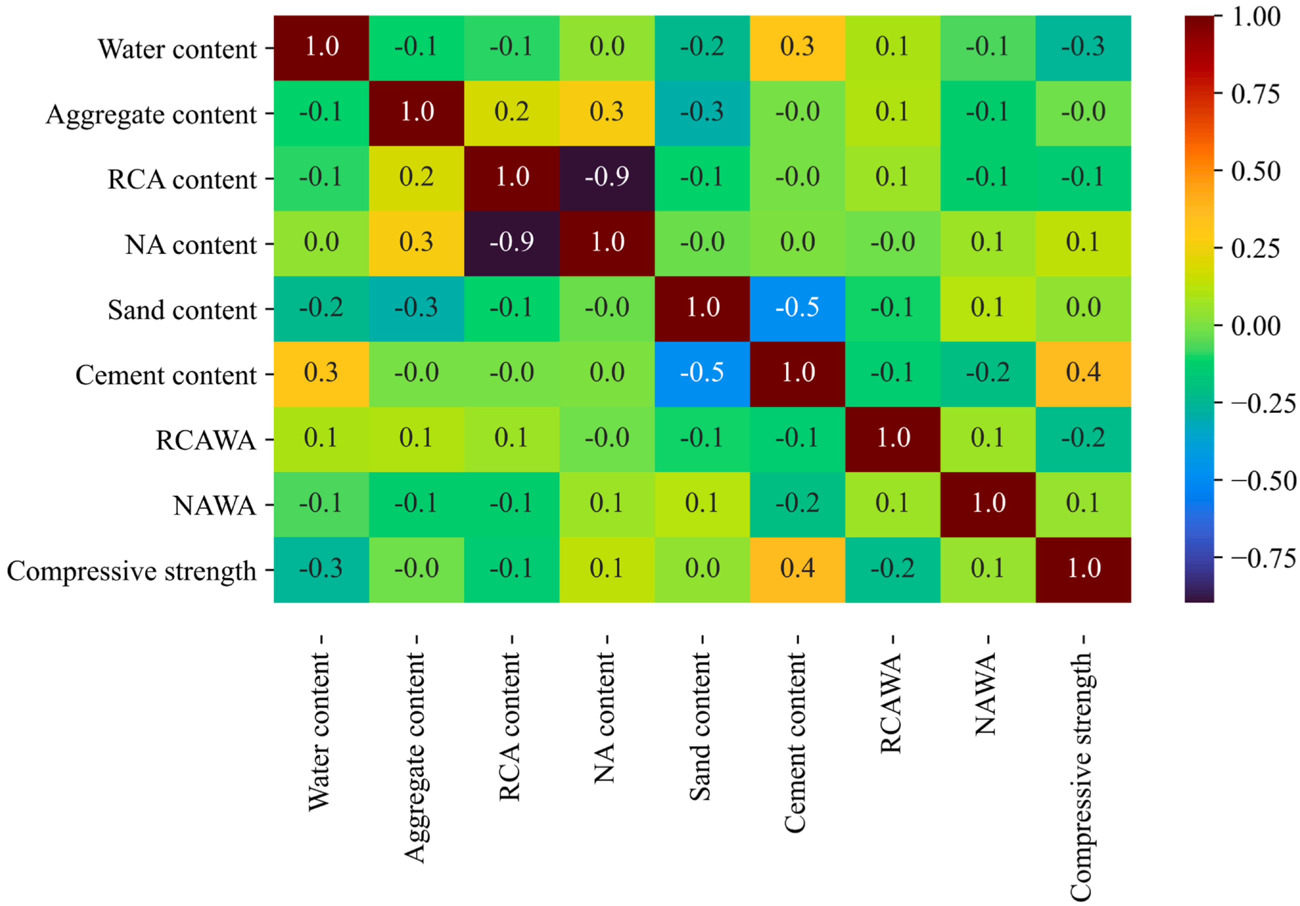


Fig. 2. Multi-correlation matrix of the input parameters and output.

aggregate content, recycled concrete aggregate (RCA) content, natural aggregate (NA) content, sand content, RCA water absorption (RCAWA), NA water absorption (NAWA). The minimum, average, median, maximum values of these input variables are described in detail in Table 2.

The effect of input variables on the compressive strength of concrete with RCA was visualized using a Hex contour graph as shown in Fig. 1. The darkish areas in this figure show the high concentration of input variables. It can be observed that the input variables have a wide range of values. Water content has a wide range from 120 ÷ 230 (kg/m^3) but mostly locates from 150 to 200 (kg/m^3) (Fig. 1a). The aggregate content is mainly concentrated in the range of 850 ÷ 120 kg/m^3 (Fig. 1b). In contrast, the RCA content has two main ranges, the first one is from 0 to around 500 (kg/m^3) with a high case without RCA content (i.e RCA content equal to 0 kg/m^3), the second one locates from 800 to approximately 1200 (kg/m^3) (Fig. 1c). Regarding the NA content, there are many cases without NA content, and a scattered number of NA content varies widely from 250 to 1300 (kg/m^3) (Fig. 1d). Fig. 1e shows the distribution of sand content, it can be seen that sand content has a wide range from 400 to 1000 kg/m^3 with some values larger than 1000 kg/m^3, the most concentrated value of NA content is 700 kg/m^3. For the case of cement content, its value largely varies from 250 to 450 kg/m^3, also there are some values greater than 500 kg/m^3 (Fig. 1f). The RCAWA mainly locates from 2.5% to 7%, and the most concentrated value is 5%, in addition, there are several values greater than 10% (Fig. 1g). In terms of NAWA, the values range from 0.5% to 2%, and there are some values higher than 2% (Fig. 1h). With all those input variables, the values of compressive strength of concrete using RCA are in the range from 10 to 70 MPa and mostly concentrate from 30 to 50 MPa.

Fig. 2 shows the multi-correlation matrix of the input parameters and output used in this study. The different colors indicate the different values of correlation. It can be observed that among the input variables, the highest correlation (R = 0.3) between them was found between water content and cement content. This is in good agreement with many previous studies because cement content and water content are well known as the most important factors affecting not only fresh properties but also hardened characteristics (such as compressive, tensile, etc.) of concrete [5-7,9]. In addition, there is a quite strong relation between aggregate content versus NA content and RCA content. Because as reported that the amount of RCA has a significant impact on the compressive strength, thus a partial replacement of NA content by RCA will lead to a change in the strength of concrete [6-8]. For the case of correlation between input variables and output variable (compressive strength), as expected, the cement content has the strongest correlation with compressive strength, followed by NA content. Overall, the correlation between input variables and output one is comparatively low. Thus, all input variables are employed to improve the accuracy of the model as well as to verify the role of the individual variable on the estimated value of compressive strength.

3. Machine learning methods

3.1. Gradient boosting (GB)

The gradient boosting (GB) was firstly proposed by Friedman [93]. It is an ensemble method whose fundamental algorithm is based on the boosting technique. In the GB algorithm, the strong learner was created by sequence addition of the weak learner to the model. The performance of the model could be consolidated based on the incorporation of various predictors from each iteration. The loss of the model or the overall error

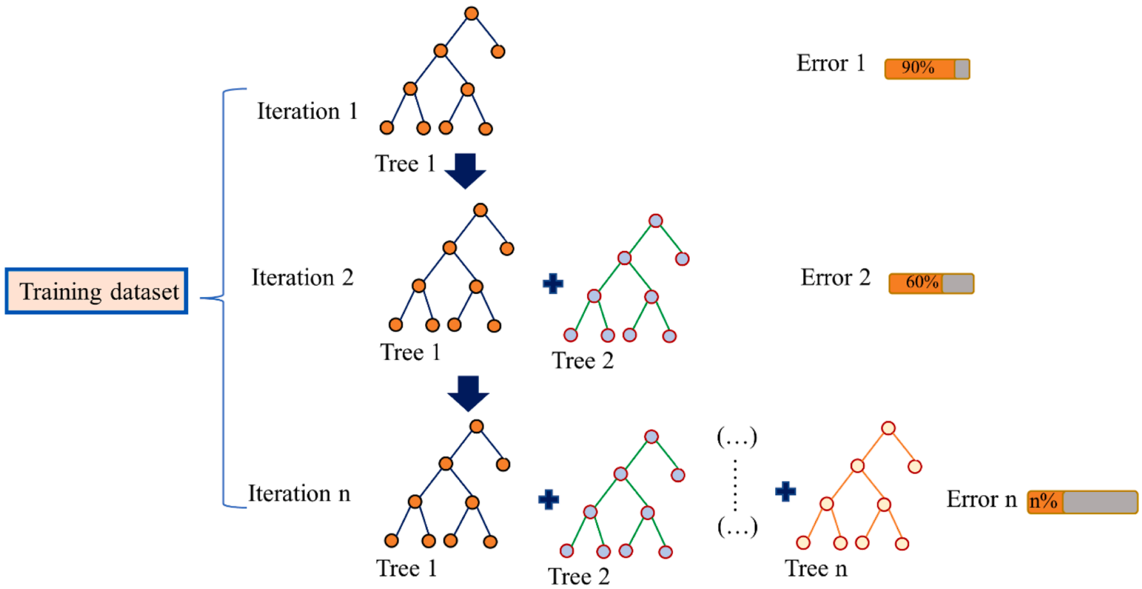


Fig. 3. The structure of the GB algorithm.

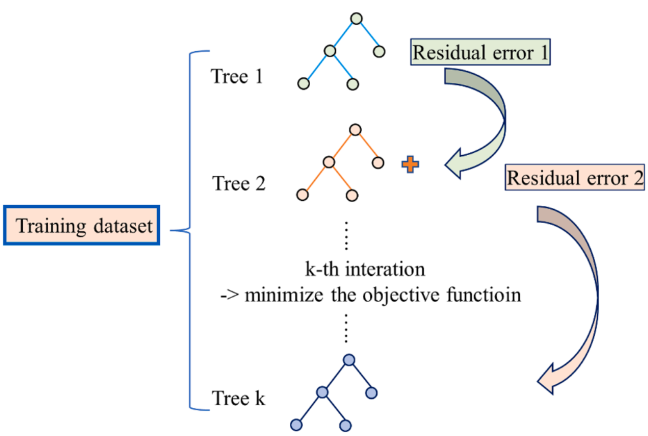


Fig. 4. The structure of the XGB algorithm.

of the model could be minimized. Hence, the overfitting due to the GB algorithm can be reduced [94]. The disadvantage of the GB algorithm may lie in the poor generalization ability when the final model was too close to the training dataset. During the GB technique, the regression tree was used as the weak learners and the stochastic gradient descent was used to train in each iteration of the model to minimize the error. Particularly, the GB algorithm learned the first weak learner (first tree) to eliminate the error in the first iteration. The GB algorithm continuously learned the second weak learner (second tree) to minimize the error from the first tree in the second iteration. The process was repeated until the desired error can be reached. Fig. 3 presents the structure of the GB algorithm.

3.2. Extreme Gradient Boosting (XGB)

Extreme gradient boosting (XGB) is firstly introduced by Chen and Guestrin [95]. It is a new ensemble technique incorporating multiple weak learners. Basically, the XGB is conducted based on the boosting algorithm where a strong learner was generated by the sequential addition of the weak learners. Particularly, in each step, a weak learner

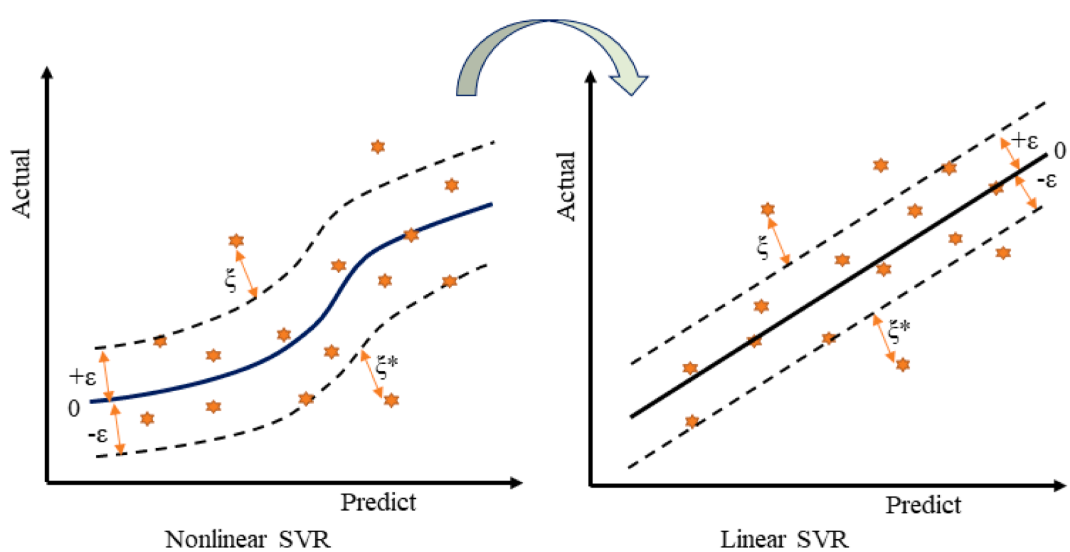


Fig. 5. Structure of SVR algorithm.

Table 3
Hyper-parameters space of machine learning (ML) models.

Gradient Boosting		Extreme Gradient Boosting		Support Vector Machine	
Number of trees	1–500	Number of trees	1–500	C	1–500
Learning rate	0.01–0.9	Learning rate	0.01–0.9		
Max features	1–8	–	–	Kernel	'sigmoid','poly','rbf'
Min samples split	0.001–0.9	–	–	Epsilon	1–100
Min samples leaf	0.001–0.9	–	–	–	–
Max depth	1–10	Max depth	1–6	–	–
Data used	10Fold CV	Data used	10Fold CV	Data used	10Fold CV
Performance index	R	Performance index	R	Performance index	R

was added to the current model to reduce the loss function of the previous model by using the gradient descent optimization algorithm. The XGB can be considered as an advantage technique of the gradient boosting algorithm because it can effectively prevent overfitting problems based on the compatible regulatory functions. The structure of XGB is shown in Fig. 4. As can be seen from Fig. 4, the strategy of the XGB algorithm was continuously added the residual error to the new tree and trained the new tree to minimize the errors until the last iteration. Thus, the superiority of the XGB algorithm is the formation of the reliable objective function for creating the tree. The optimal value of the objective function can be given as follows:

$$Obj = \gamma T + \sum_{i=1}^{T}\left[g_i w_i + \frac{1}{2}(h_i + \lambda)w_i^2\right] \tag{1}$$

where λ and γ are the regular factors used to avoid the overfitting; T is the total number of tree leaves; g_i and h_i are the first and second-order derivations, respectively; w_i is the leaf weight.

3.3. Support vector machine regression (SVR)

The classification problems could be solved through the support vector machine algorithm that was proposed by Vapnik [96]. The support vector machine was later extended and called as support vector machine regression (SVR) for the solution of regression and prediction problems [97]. Based on the structural risk minimization principle, the SVR algorithm attempted to minimize the generalization error and obtain the optimal solution. In this study, the Gaussian radial basis function was employed to transform from a lower-dimensional feature space into a higher one so that it was possible to perform the linear regression and better in fitting the training dataset, as shown in Fig. 5. The output variable can be given as a linear function as follows:

$$f(x) = \langle \omega \bullet \varphi(x) \rangle + b \tag{2}$$

where $\langle \bullet \rangle$ indicates the dot function; $\varphi(x)$ is a transformation that maps the input feature to a higher dimensional feature space; b is the parameter vector of the function (model bias); ω are the minimum values which are derived by the following equation:

$$minimise\left[\frac{1}{2}\|\omega\|^2 + C\sum_{j-i}^{n}\left(\xi_j + \xi_j^*\right)\right] \tag{3}$$

$$subjectto\left\{ y_j - \omega \bullet \varphi(x_j) - b \le \varepsilon + \xi_j \omega \bullet \varphi(x_j) + b - y_j \le \varepsilon + \xi_j^* \xi_j, \xi_j^* \ge 0\right.$$

where ξ_j, ξ_j^* are the slack variables; C is the regularization parameter; ε is the insensitive loss function (error between the test and predicted values); y_j is the test value; n is the number of samples.

3.4. Partial dependence plots

Partial dependence plots (PDP) is a common technique that was proposed by Friedman for visualizing the ML output [93]. By PDP results, the partial relationship between the predicted values and input features can be identified, i.e., linear, monotonic, or complex. In other words, the PDP can illustrate how the predicted value could vary according to one or more input features. The average partial dependence function, $\overline{F_S}$, on a subset of predictors k_S is measured as follows:

$$\overline{F_S}(k_S) = \int \widehat{F}(k_S, k_C)dP(k_C) \tag{4}$$

where k_S is the target feature of the input variables; k_C is the complement such that $k_C \cup k_S = S$; $dP(k_C)$ is the marginal effect of k_C .

For training data $\{S_i, i = 1, 2, \cdots, n\}$, the average partial dependence function in Eq. (4) can be constructed as follows:

$$\overline{F_S}(z_S) = \frac{1}{N}\sum_{i=1}^{N}\overline{F}(k_S, k_{i,C}) \tag{5}$$

where N is the total number of samples; $k_{i,C}$ is the real value of the i-th variable in the training dataset. Rather than only plot, the effect of one input parameter on the target feature, an individual conditional expectation plot (PDP-1D) introduced by Goldstein et al. [98] was conducted in this study. It insightfully visualized the marginal effect of a given input feature on the predicted output. Besides, two-dimension partial dependence plots (PDPs-2D) were also employed to obviously present the interaction between the predicted output and two input parameters.

3.5. Particle swarm optimization (PSO)

Particle swarm optimization (PSO) is a global optimization method that is applied in the machine learning model to adjust the hyper-parameter. The PSO algorithm was inspired by the behavior of bird swarming and fish training [99]. Ebeltagi et al. [100] concluded that the PSO was the best technique by comparing five optimization algorithms, i.e., PSO, ant colony systems, genetic algorithm, shuffled frog leaping, and memetic algorithm. In the PSO, the potential solution known as a particle was randomly initialized in the searching range. The global best solution was then searched by iteratively updating the swarm of particles. The particle direction can be modified based on the best position of previous and other particles as follows:

$$X_{id}^{(t+1)} = X_{id}^{(t)} + V_{id}^{(t+1)} \tag{6}$$

$$V_{id}^{(t+1)} = \omega \times V_{id}^{(t)} + C_1 \times R_1 \times \left(P_{id} - X_{id}^{(t)}\right) + C_2 \times R_2 \times \left(P_{gd} - X_{id}^{(t)}\right) \tag{7}$$

where $X_{id}^{(t)}$ and $V_{id}^{(t)}$ are the location and velocity of "*i-th*" particle at the "*t-th*" iteration, respectively; $X_{id}^{(t+1)}$ and $V_{id}^{(t+1)}$ are the location and velocity of "*i-th*" particle at the "*(t + 1)-th*" iteration, respectively; ω is the initial factor; C_1 and C_2 are acceleration constants in [0,4]; R_1 and R_2 are the random function in [0,1]; P_{id} is the best position of an individual particle; P_{gd} is the best global position of the entire swarm. Table 3 shows

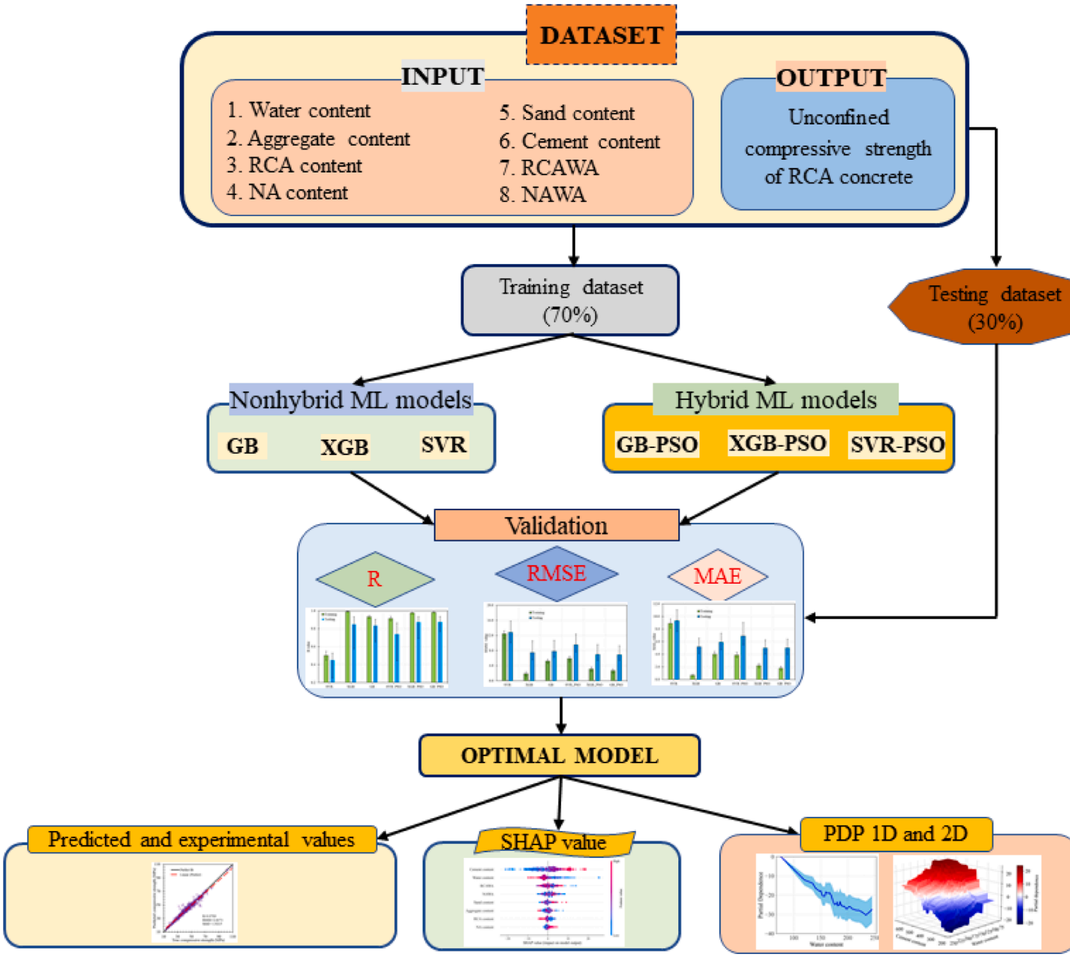


Fig. 6. The methodology flowchart of the ML model in this study.

the application of the PSO algorithm in this study to find out the best hyper-parameter for the machine learning model.

3.6. Performance evaluation of models

After building the predicted model, it is necessary to evaluate its performance. Hence, three statistical parameters were employed to assess the accuracy of the proposed model in this study, i.e., root mean square error (RMSE), mean absolute error (MAE), and correlation coefficient (R). Although both RMSE and MAE explicitly measure the magnitude of residual error [101], the best model was primarily chosen from the RMSE results [102]. The lower values of RMSE and MAE, the better the forecasting model. Meanwhile, R indicates the different degrees between the actual and predicted values and varies from 0 to 1. The higher R-value implies the better-proposed model. The formulations of three statistical indexes are written as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{\mathrm{j}=1}^{N}\left(p_{0,\mathrm{j}}\text{-}\mathrm{p}_{\mathrm{t,j}}\right)^2} \tag{8}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{\mathrm{j}=1}^{N}\left|p_{0,\mathrm{j}}\text{-}\mathrm{p}_{\mathrm{t,j}}\right| \tag{9}$$

$$R = \frac{\sum_{j=1}^{N}\left(p_{0,j}-\overline{p_0}\right)\left(p_{t,j}-\overline{p_t}\right)}{\sqrt{\sum_{j=1}^{N}\left(p_{0,j}-\overline{p_0}\right)^2\sum_{j=1}^{N}\left(p_{t,j}-\overline{p_t}\right)^2}} \tag{10}$$

where p0,j is the compressive strength compliance of i-th actual value in the dataset; Pt,I is the compressive strength of i-th predicted value obtained from ML model; P0is the mean actual value of the compressive strength compliance; pTis the mean predicted value of the compressive strength; N is the total number of samples in the dataset. Moreover, Saberi-Movahed et al. [103] recommended that the reliability of the ML model should be carried out to assess its consistency. Therefore, the reliability of the proposed ML model was also performed in this study. More detail of the implementation can be found in the existing study [103].

3.7. Methodology flow chart

Fig. 6 shows the flowchart of ML model application for predicting the compressive strength of RCA concrete in this study. Generally, three are four main steps as follows:

Step 1. Preparing Dataset:

All input factors including water content, aggregate content, RCA content, NA content, sand content, cement content, RCA water absorption, NA water absorption, and compressive strength of RCA concrete were collected to build the ML models in this step. The total dataset comprises complete 721 RCA concrete samples which were obtained from the previous studies. Then, the dataset was randomly split into 2 groups, i.e., 70% of the dataset (training dataset) was used for training the ML model and 30% of the dataset (testing dataset) was used for checking the performance of the ML model.

Step 2. Training the proposed ML models:

Table 4
Optimum hyper-parameters of ML models.

Gradient Boosting		Extreme Gradient Boosting		Support Vector Regresion	
Number of trees	151	Number of trees	1–500	C	200
Learning rate	0.124	Learning rate	0.1		
Max features	3	–	–	Kernel	'rbf'
Min samples split	0.041	–	–	Epsilon	2
Min samples leaf	0.0167	–	–	–	–
Max depth	6	Max depth	3	–	–
Performance index	0.8819	Performance index	0.8741	Performance index	0.7433

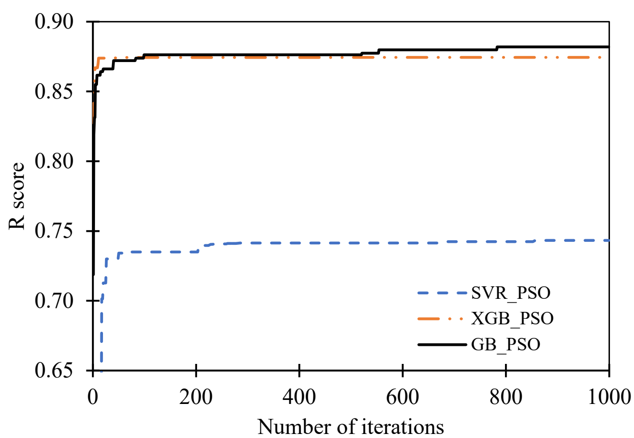


Fig. 7. R mean value versus a number of iterations using hybrid models and 10-Fold cross-validation.

The training dataset was used to train all proposed ML models in this step. Three algorithms were applied to train three ML models, i.e., GB, XGB, and SVR. Besides, the PSO algorithm was conducted to consolidate the models. Specifically, the ML models in this study included two groups. The first group included the individual ML models and the second one was the combination of PSO and ML models. When a tolerance error criterion was obtained, the best model was defined.

Step 3. Validating the ML models:

The testing dataset was used in this step to validate the ML models in step 2. All ML models were then checked through three statistical parameters including RMSE, MAE, and R.

Step 4. Evaluating the input parameters and comparing the ML models:

The feature importance using SHAP value was employed to assess the most importance and influence of the input parameters on the compressive strength of RCA concrete. Besides, the sensitivity analysis using PDP-1D and PDP-2D was carried out to evaluate the capability for prediction of the compressive strength based on one or two input parameters. Finally, the results obtained from predicted ML models were compared with the actual experimental results.

4. Results and discussion

4.1. Performance evaluation of hybrid models

The optimum hyper-parameters of the used ML moded in this study are presented in Table 4. The trial–error process was implemented to achieve the optimal values, which correspond to the highest prediction accuracy of the proposed models. In the process to achieve the optimal value, each parameter changed in a specific range, while other parameters were kept as default values. After that, the predictive accuracy of each model is computed, then the fine-tuned process was conducted to achieve the corresponding values. The more detailed information of these parameters for the trial–error test can be referred to the previous section (see Table 3). From Table 4, it can be seen that the Gradient Boosting has the highest performance index of 0.8819, followed by Extreme Gradient Boosting, and Support Vector Regression. The detailed information related to the optimal para terms of each technique also can be found in Table 4.

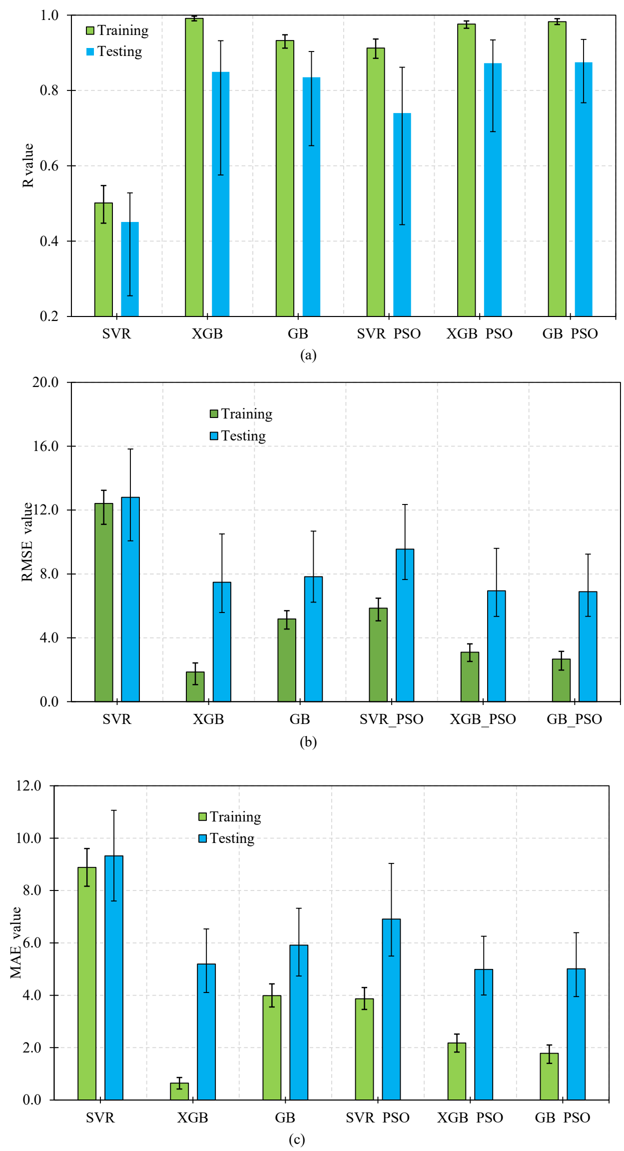


Fig. 8. Performance of machine learning models after 3000 simulations.

The R mean value of hybrid models versus the number of interactions using 10-Fold cross-validation is shown in Fig. 7. Generally, all three hybrid models have a high R score (lager than 0.73) with a range of number interactions from 50 to 1000. For all number of iterations, the R scores of GB_PSO and XGB_PSO were almost equal, the R score of GB_PSO was slightly higher than that of XGB_PSO. The R score of SVR_PSO was the lowest and there is a large gap of the R score between SVR_PSO compared to the two remaining algorithms. The R score

Table 5
Comparison of six algorithms for compressive strength prediction using mean performance value.

Algorithm	Training set			Test set		
	R	RMSE/ MPa	MAE/ MPa	R	RMSE/ MPa	MAE/ MPa
SVR	0.502	12.420	8.885	0.451	12.798	9.325
XGB	0.991	1.855	0.645	0.850	7.478	5.193
GB	0.932	5.181	3.988	0.835	7.828	5.916
SVR_PSO	0.913	5.856	3.868	0.740	9.553	6.913
XGB_PSO	0.976	3.098	2.182	0.872	6.946	4.989
GB_PSO	**0.983**	**2.666**	**1.782**	**0.875**	**6.890**	**5.008**

achieved a constant value when the number of iterations was larger than 200 for SVR_PSO and XGB_PSO, while the GB_PSO attained a constant value of R when the number of iterations was greater than 800.

The performance of six machine learning models using 3000 simulations is shown in Fig. 8. The R-value of six algorithms is shown in Fig. 8a. The R-value of the training set is always higher than that of the testing set for all algorithms. In addition, the standard deviation of the training set was smaller than that of the testing set for all models. Among all models, the model using SVR has the lowest R score with the highest standard deviation. With regard to RMSE and MAE, the highest values of RMSE and MEA were obtained in the model using SVR (Fig. 8b and 8c), this means that SVR has the lowest accuracy, and these results agreed well with the result of R score. Similar to the R score case, the standard deviation of the testing part is always higher than that of the training part for all algorithms. The XGB model has the lowest value of RMSE and MAE for the training part. Overall, the hybrid models of GB_PSO have the best performance with a high R score, low values of RMSE and MAE with the smallest standard deviation.

The comparison of the compressive strength prediction of six algorithms using mean performance values is shown in Table 4. In general, a higher value of R and a smaller value of RMSE and MAE indicates that the model has a higher and better prediction ability. Regarding the R-value for the training set, all algorithms performed very well with R-value larger than 0.900 except the SVR algorithm. Similar to the training set, a high value of R (lager than 0.700) was found for all algorithms except SVR. Among three single algorithms, XGB has the highest prediction ability, and the lowest was found for the case of SVR. In terms of hybrid models, the GB_PSO shows the highest prediction ability with a very high value of R and low values of RMSE and MAE for both training and testing sets, followed by XGB_PSO and SVR_PSO. This means that incorporated with PSO, the prediction of the GB_PSO was improved and better than the single GB model.

Table 5 shows the comparison of six algorithms for the prediction of compressive strength using the standard deviation of performance value. All algorithms performed well with a very small value of standard deviation, which indicates high accuracy of prediction. Overall, the accuracies of hybrid models are higher than those of single models. This can be explained by the advantages of PSO, which could optimize the problem and enhance the prediction ability of the hybrid model as reported in previous studies [25,26]. Among these six algorithms, GB_PSO has the lowest standard deviation, this indicates that GB_PSO has the highest prediction ability.

Based on the results of the comparison in Tables 4 and 5, it can be concluded that the performances of hybrid models are better than those of single models. This is because PSO has some advantages concerning evolutionary algorithms [25,26], for instance, PSO has no complicated operators as evolutionary algorithms as it has fewer parameters that need to be adjusted [24]. Thus, PSO could enhance the performance of the single algorithm such as it can improve the speed of modeling. As a result, the hybrid models are employed for modeling to predict the compressive strength of recycled concrete. Fig. 9 shows the comparison of predicted and actual values of compressive strength for both training and setting parts. The comparisons between predicted and actual values of compressive strength using GB_PSO for both training and testing sets are shown in Fig. 9a and 9b. There is a very close relationship between predicted and actual values with correlations R = 0.9789 and R = 0.9356 for training and testing sets, respectively. The training set has a higher value of R and lower values of RMSE and MAE than those of the testing set. Similarly, for the case of the XGB_PSO model, the training set has a better performance than the testing set. The values of R are 0.9728 and 0.9342 for the training and testing set, respectively, while the values of RMSE are 3.2524 MPa and 8.1984 MPa and MAE are 2.2642 MPa and 6.0073 MPa for the training and testing set, respectively (Fig. 9c and 9d). The lowest prediction accuracy is found for the SVR_PSO for both the training and testing set (Fig. 9e and 9f). The R-values are 0.8969 and 0.8617, respectively; the values of RMSE and MAE are slightly higher than those found in the XGB_PSO model. The results of Fig. 9 indicate that the GB_PSO produced the highest prediction accuracy among three hybrid models. This result is in agreement with the result of the previous study [20]. In their study, they revealed that the GB performed better

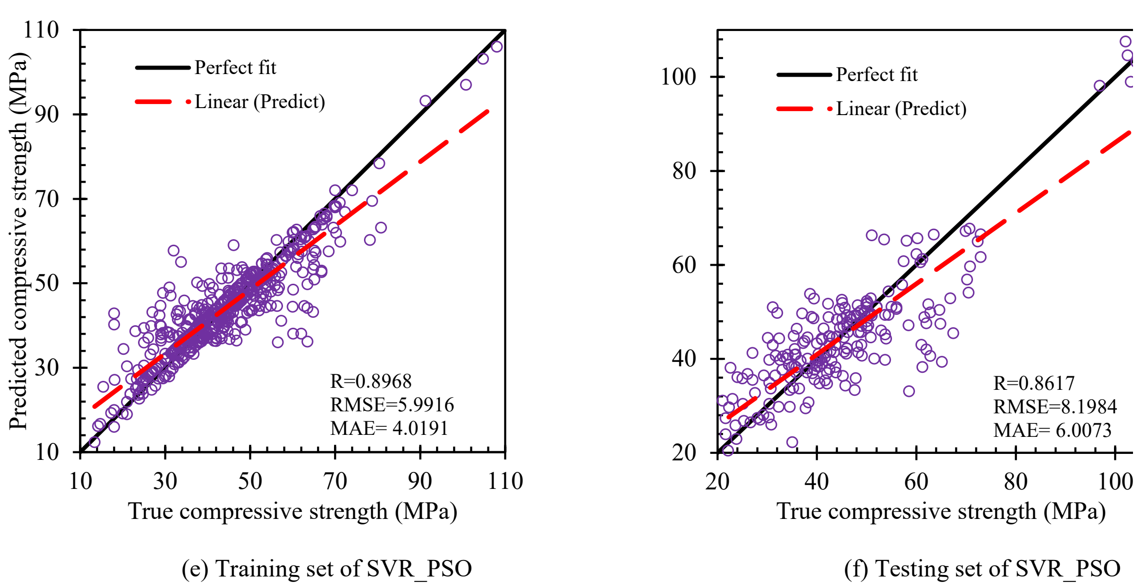


Fig. 9. Comparison of predicted compressive strength and actual values using train and test data with best case of each hybrid model.

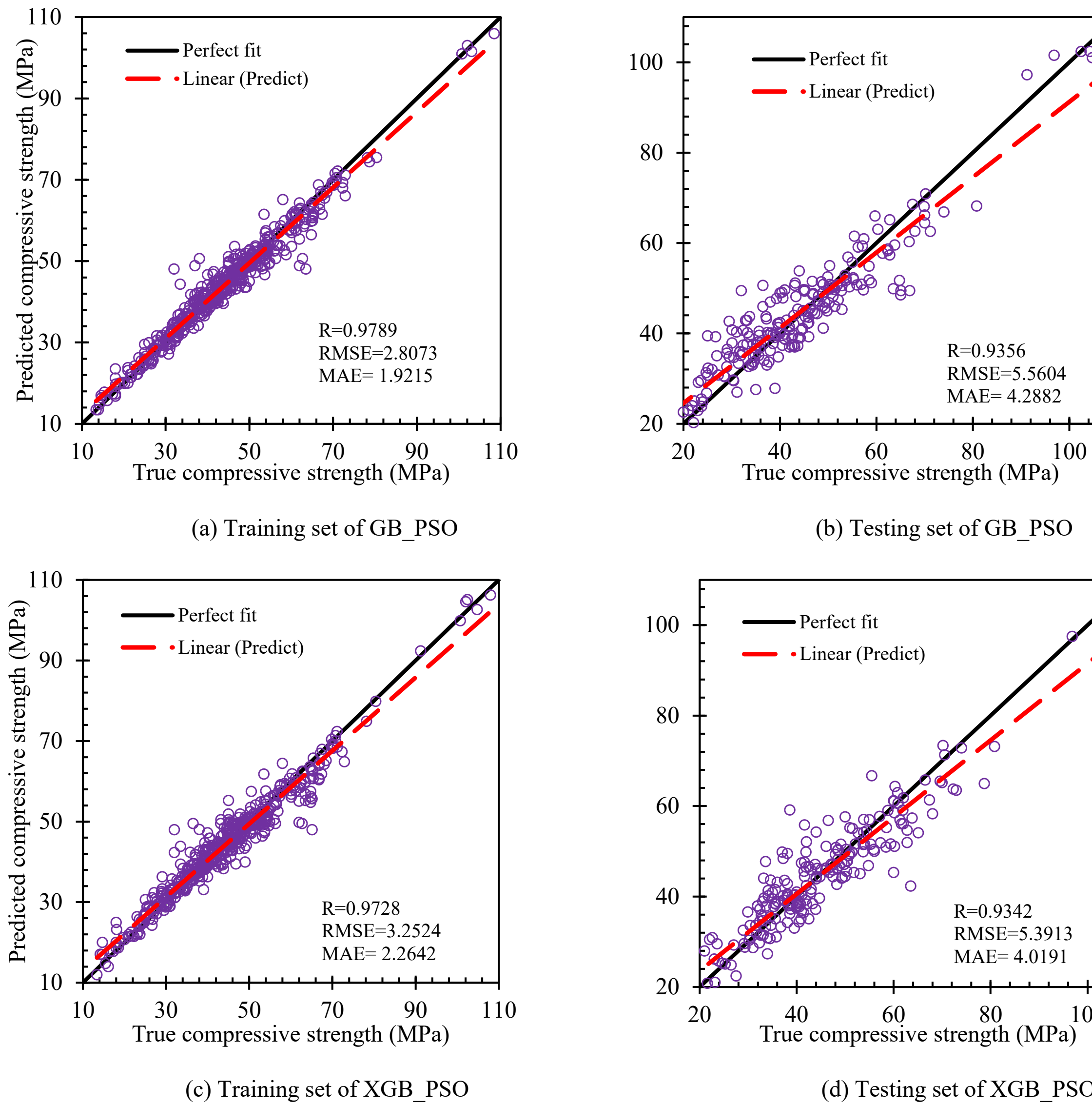


Fig. 9. (*continued*).

Table 6
Comparison of six algorithms for compressive prediction using the standard deviation of performance value.

Algorithm	Training set			Test set		
	R	RMSE/MPa	MAE/MPa	R	RMSE/MPa	MAE/MPa
SVR	0.015	0.297	0.204	0.031	0.855	0.517
XGB	0.002	0.236	0.072	0.037	0.675	0.369
GB	0.005	0.168	0.132	0.030	0.516	0.349
SVR_PSO	0.007	0.201	0.118	0.055	0.673	0.425
XGB_PSO	0.003	0.154	0.094	0.028	0.558	0.332
GB_PSO	**0.002**	**0.165**	**0.088**	**0.024**	**0.501**	**0.328**

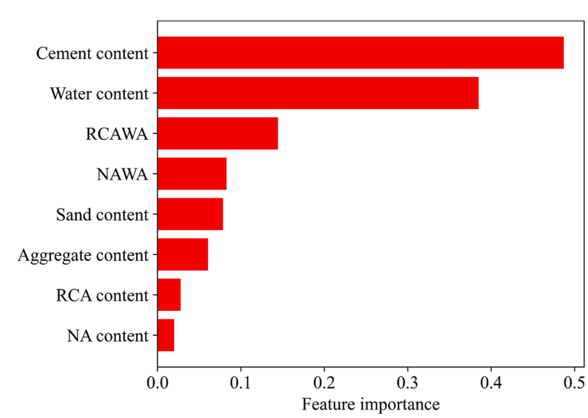


Fig. 10. Feature importance analysis using the Sklearn_ permutation_importance library in Python code.

than that of the Gaussian process and deep learning.

Table 6 presents a comparison of statistical indexes in this study and other previous works. From this table, we can see that this study used three indicators for both training and testing sets, however, for other previous studies, only one or two indicators were used for modeling, and the results mainly came from the testing set. Thus, it can be indicated that this study provided systematic information on modeling. The result

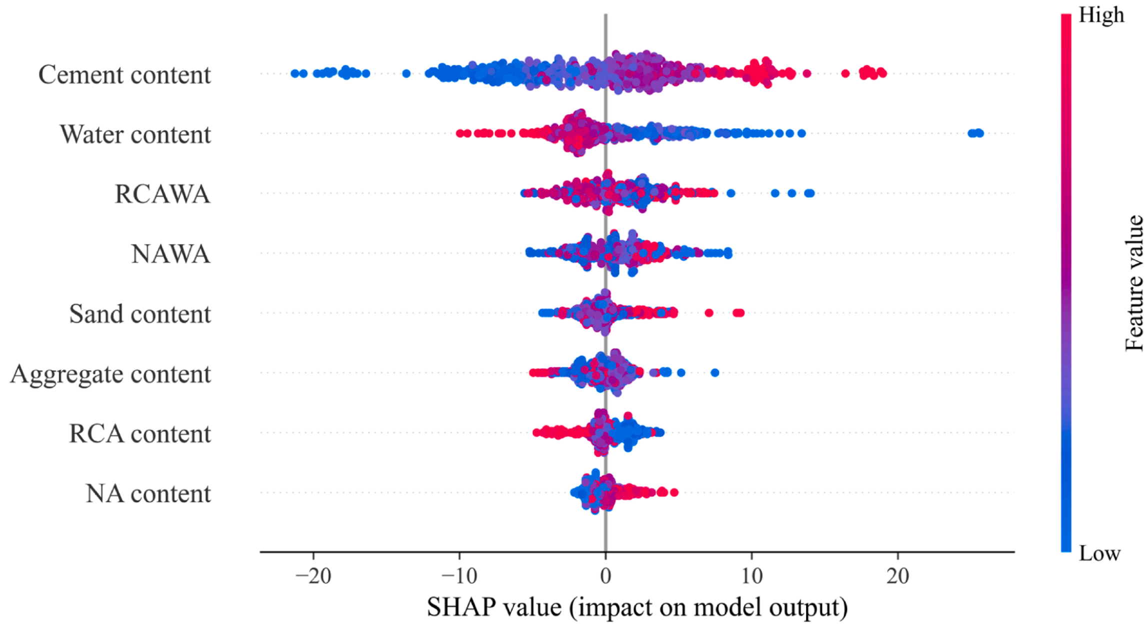


Fig. 11. Feature importance analysis using the SHAP library in Python code.

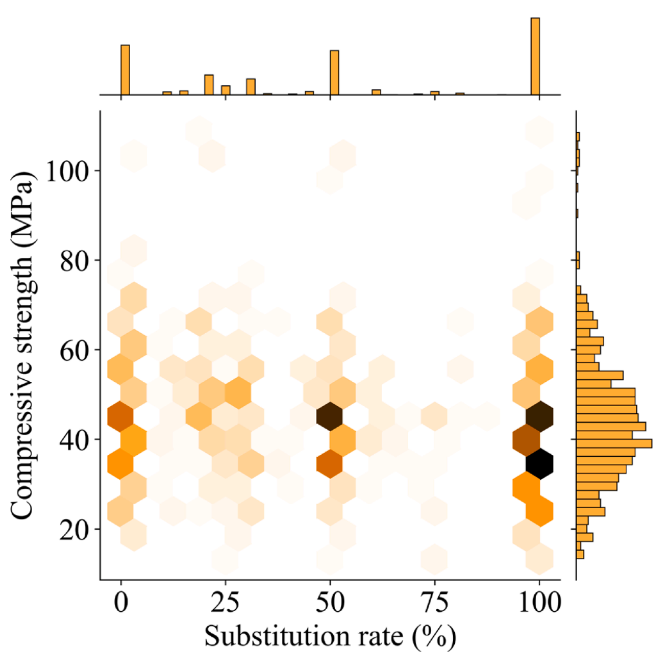


Fig. 12. Hex contour including correlation and histogram distribution of compressive strength and substitution rate of RCA.

of the GB_PSO model in this study achieved better accuracy than that of the previous study using five single machine models (see detail in Table 6) conducted by Gholampour et al. [104], which has a similar number of data samples (650 samples). The result of the GB_PSO model has higher prediction accuracy than that of Naderpour et al. [105] using ANN (139 data samples) and Deshpande et al. [3] using model tree and non-linear regression model (257 samples) except model using ANN, but lower than that of Duan et al. [4] using ANN (168 data samples) and Topçu and Saridemir [18] (210 data samples). This means that although higher data samples might produce a better generalization of the model, the prediction accuracy can be lower. Because accuracy metrics alone may not be sufficient to evaluate the prediction performance of the model and another indicator such as feature importance is also crucial to assess the predictive accuracy [20]. Moreover, it can be observed that the result of the GB_PSO model in this study has almost similar prediction accuracy compared to the results in the previous study using GB (1134 data samples) [20], or slightly lower than the results of another previous study using ANN (1178 data samples) [17]. This indicates that the PSO algorithm has improved the prediction ability of the hybrid model of GB_PSO even the numbers of data samples in this study are smaller than those in previous studies. Nevertheless, in the previous study by Dantas et al. [17], the R-value (0.963) of the training set was smaller than that (0.985) of the testing set, this means that the model in the previous study using ANN was not trained appropriately or sufficiently as stated by Gulli and Pal [106].

4.2. Feature importance analysis

The dependence of the compressive of concrete using RCA for each input variable is shown in Fig. 10. From the figure, it can be observed that cement content is known as the most important variable affecting compressive strength. This is in alignment with the practical knowledge of the concrete. The second important variable is obtained for the case of water content, and this agreed well with the common practice of concrete. The amount of RCAWA is known as the third important input variable. From this figure, we can also see that the NA content is the least important variable and RCA content is more important than NA content.

It is important to know how a proposed algorithm analyzes as well as computes the detail process to predict the compressive strength. To clarify the prediction and importance ranking of each input, a game theory namely SHAP was adopted to compute the contribution of each input [107]. The value of SHAP is the average marginal influence of each input variable through all potential combinations of input variables. Based on the absolute value of SHAP, the importance of each variable can be estimated. The variables having a larger absolute value of SHAP are known as crucial variables. To obtain the global feature importance, the absolute value of SHAP for all variables is computed as the mean value, and then the absolute values of SHAP were arranged in order from large to small (from right to left in the x-axis). The SHAP value of each variable is presented as each point on the graph. The x-axis indicates the SHAP value of each variable, while the y-axis indicates the feature importance.

The feature importance analysis of the compressive strength of

Table 7
Comparison of statistical indexes with previously published research.

Name of technique	Training R	RMSE	MAE	Testing R	RMSE	MAE	Total number of samples	Reference
SVR_PSO*	0.897	5.992	4.019	0.862	8.198	6.007	721	this study
XGB_PSO*	0.973	3.252	2.264	0.934	5.391	4.019		
GB_PSO*	**0.979**	**2.807**	**1.922**	**0.936**	**5.560**	**4.288**		
GB	–	–	–	0.959	5.076	–	1134	[20]
Deep learning	–	–	–	0.932	6.502	–		
ANN	0.963	–	–	0.985	–	–	1178	[17]
ANN	0.903	–	–	0.829	–	–	139	[105]
ANN	0.999	2.047	–	0.999	2.395	–	210	[18]
Fuzzy logic (FL)	0.999	2.656	–	0.998	3.866	–		
Multivariate adaptive regression splines	–	–	–	–	9.100	5.400	650	[104]
M5 tree	–	–		–	8.300	5.900		
Least support vector regression	–	–	–	–	7.700	4.600		
ANN	–	–	–	0.950	–	9.310	257	[3]
Model tree	–	–	–	0.870	–	12.020		
Non-linear regression model	–	–	–	0.860	–	14.710		
ANN	0.999	1.796	–	0.997	3.680	–	168	[4]

Note: "*" indicates model conducted in this study, "-" denotes not available.

Table 8
Description of RCA substitution rate value.

Unit	Mean	Min	$Q_{25\%}$	Median	$Q_{75\%}$	Max	Std	Skw
%	50.513	0	20	50	100	100	38.792	0.127

value of the testing part for case (iii) is the lowest in the three cases.

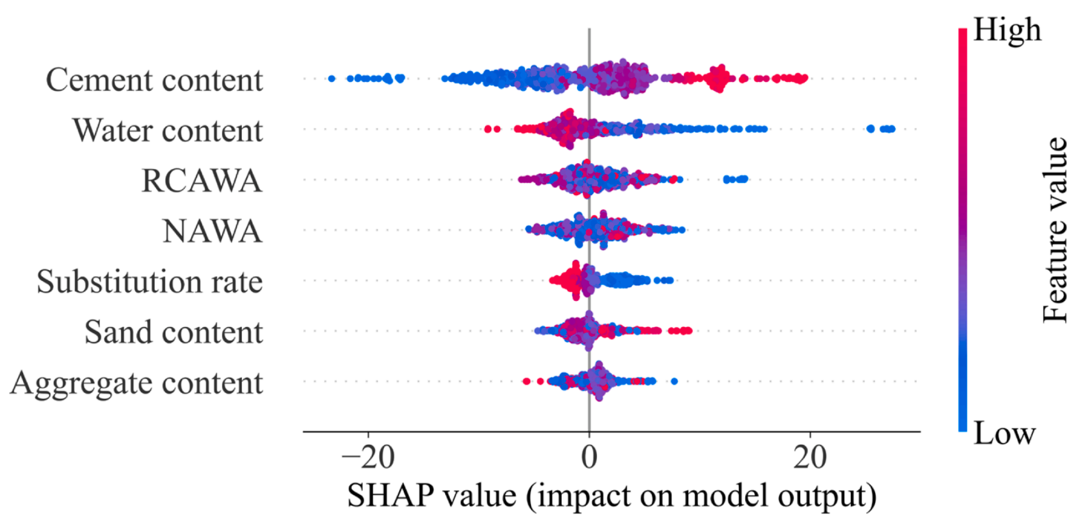


Fig. 13. Feature importance analysis using the SHAP Explanation in case (iii).

recycled concrete prediction is shown in Fig. 11. It can be seen that the most crucial input variable that greatly influences the compressive strength of concrete with RCA is the cement content, while the least influence variable is the NA content. The result of the SHAP value is consistent with the result of feature importance analysis using the Sklearn_permuation_imortance in Fig. 10. The dot with red color indicates a large feature value corresponding to a larger value of SHAP. Regarding cement content, we can see that the higher cement content leads to a higher value of SHAP, and it is agreed very well with many results of published experimental results. The second important variable significantly affects the compressive strength is the water content. As presented in the figure, the lower water content leads to higher feature importance, and the higher water content results in the lower importance of the variable. The result of SHAP is in alignment with the result of many previous studies in recycled concrete [108-111]. From these results of SHAP, it can be indicated that using the game theory approach for SHAP could enhance the understandability of the proposed hybrid machine learning models and also demonstrate that prediction accuracies of the model are reasonable and acceptable. Thus, it can be inferred that this game theory direction can increase the predictive ability and workability of machine learning models.

It is worth noting that RCA content and NA content have the lowest effect on the prediction capability and magnitude value of compressive strength. Moreover, the aggregate content is one of the input variables therefore it is interesting to introduce the substitution rate of RCA content as a new input variable replacing the RCA and NA content in the GB_PSO model with the tuned hyperparameters. The correlation between substitution rate and compressive strength, the description range value of substitution rate is shown in hex contour Fig. 12 and Table 7, respectively. The performance of the GB_PSO model is investigated in three different cases of variable (i) initial input variables, (ii) initial input variables + substitution rate of RCA, and (iii) initial input

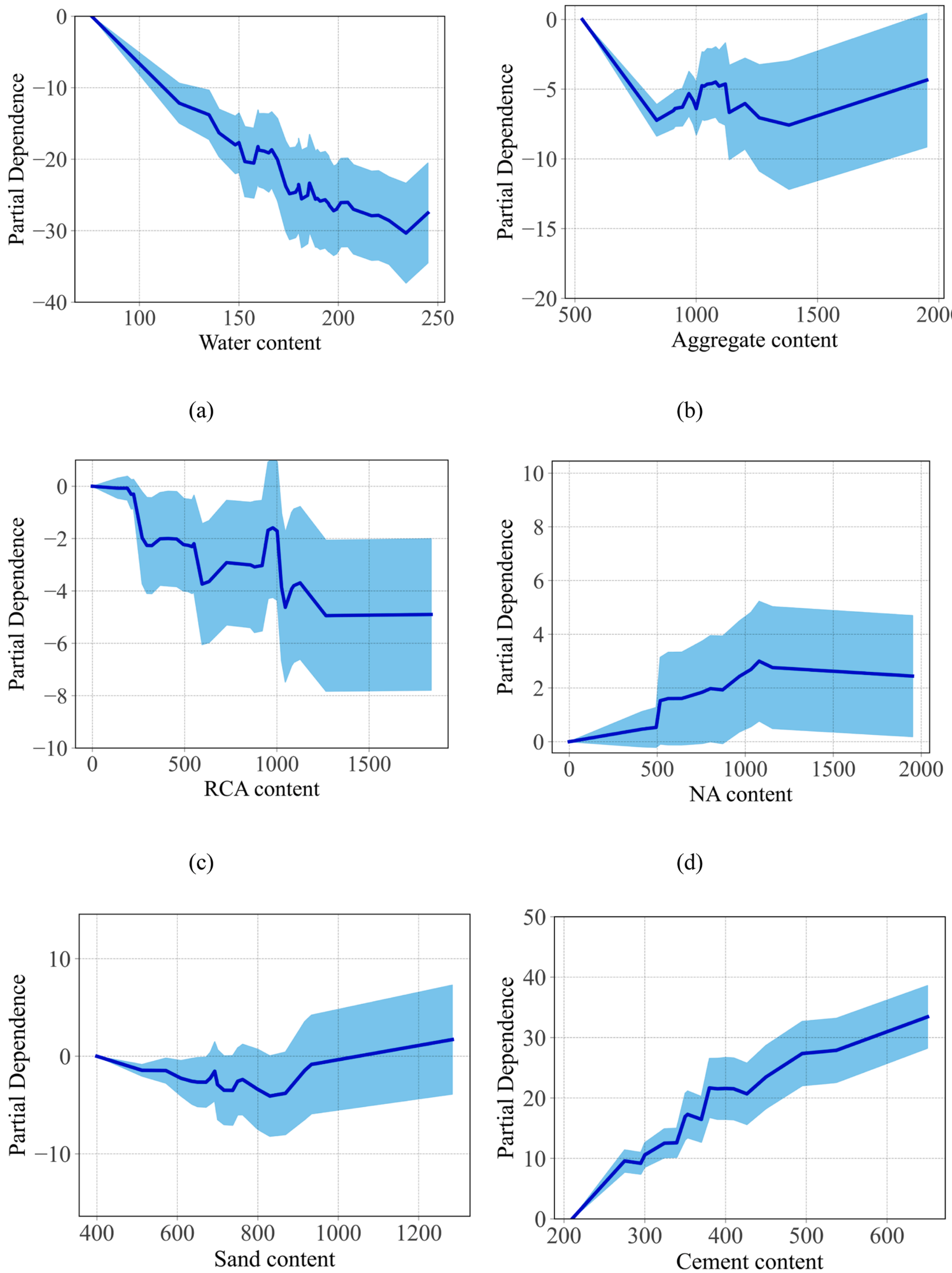
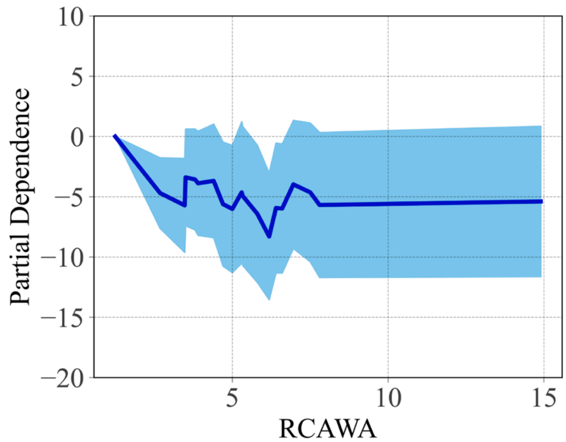


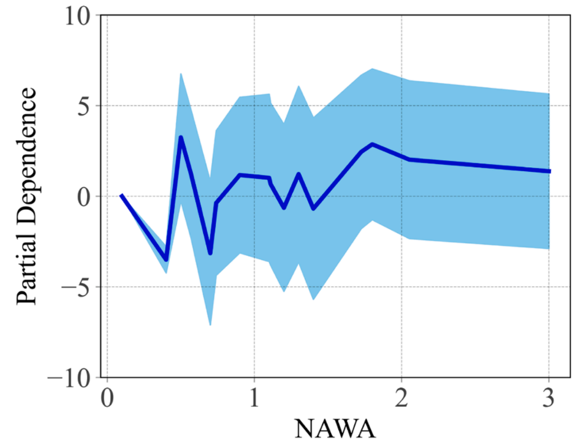
Fig. 14. Partial dependence plots (PDP) analysis 1D of the input variables for compressive strength of concrete made with recycled concrete aggregates (a) water content, (b) aggregate content, (c) RCA content, (d) NA content, (e) sand content, (f) cement content, (g) RCAWA, (h) NAWA and (i) Substitution rate (%).

variables – (RCA content, NA content variables) + substitution rate of RCA. A comparison of the performances is summarized in Table 8.

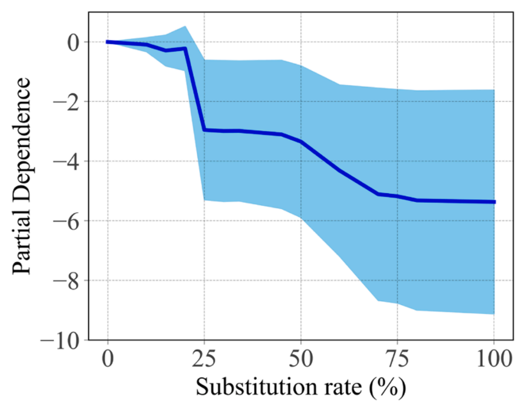
The comparison shows that if the GB_PSO model uses simultaneously supplementary new input variable as substitution rate of RCA and the initial input variables, the performance of this case is lowest. In (iii), if the GB_PSO model considers the presence of "substitution rate of RCA" and absence of RCA, NA content in the space of input variable, the performance of this case (iii) is relatively lower than that of the initial case. In fact, MAE Using the GB_PSO model of the case (iii) for global interpretation by SHAP Explanation (cf. Fig. 13), the result shows that the substitution rate of RCA has a higher effect on compressive strength of concrete than sand content and aggregate content. The range of effect will be quantified by Partial dependence analysis in the next section.

(e) (f)





(g) (h)



(i)

Fig. 14. (*continued*).

Table 9
Comparison of performance evaluation between three cases of variable.

Case	Performance evaluation								
	R_train	R_test	R_all	RMSE_train	RMSE_test	RMSE_all	MAE_train	MAE_test	MAE_all
i	0.9789	0.9356	0.9635	2.8073	5.5604	3.8490	1.9215	4.2882	2.6338
ii	0.9795	0.9310	0.9623	2.7679	5.7464	3.9107	1.8585	4.3580	2.6108
iii	**0.9786**	**0.9331**	**0.9624**	**2.8264**	**5.6636**	**3.9037**	**1.9450**	**4.2258**	**2.6315**

Table 10
Application PDP 1D in predicting compressive strength of concrete made with different values of RCA substitution rate.

	True mix design					Gain value according PDP plot				
Water content	184.9	184	114	171.5	184.8	−27	−27	−10	−24	−27
Aggregate content	1290	1104	900	967.26	1267.2	−7.5	−5	−6	−6	−7.5
Sand content	558	708	600	686	528	−1	−3	−2	−2	−2
Substitution rate	0	100	100	30	75	0	−5.5	−5.5	−3	−5.2
Cement content	430	460	600	343	528	20	27	31	12	29
RCAWA	0	3.47	7.8	1.93	14.9	0	−4	−5	−2	−5
NAWA	0.4	0	0	1.63	1.2	0	0	0	1	1
	True compressive strength					Estimated = mean value (44.15) + $\sum Gain$				
Compressive strength (MPa)	26.9	29.5	62.8	24.6	45.1	28.65	26.65	46.65	20.15	27.45

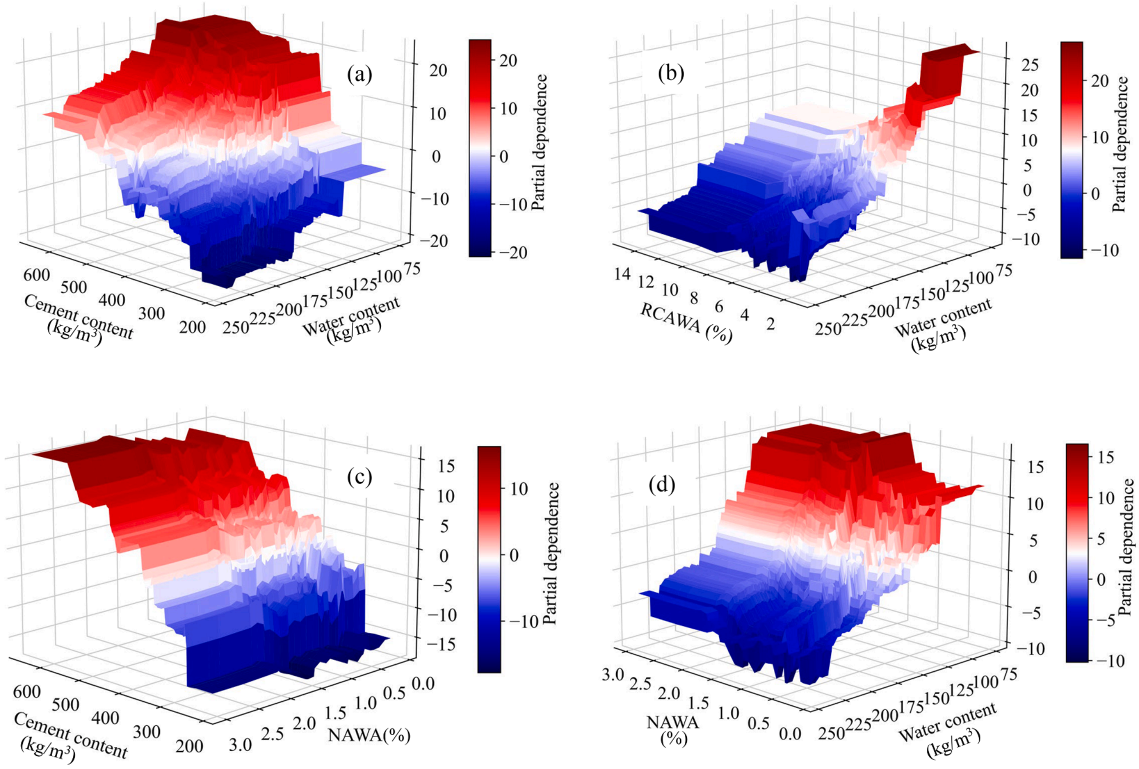


Fig. 15. Partial dependence plots (PDP) analysis 2D of the input variables for compressive strength of concrete made with recycled concrete aggregates.

4.3. Partial dependence analysis

To assess the influence of input variables on the compressive strength of the concrete with RCA, the partial dependence plots 1D (PDP) analysis was conducted and shown in Fig. 14. Indeed, one of the important advantages of PDP is that the quantity of each affected input variable can be calculated easily from the results of PDP. In the PDP analysis, the other remaining variables were kept constant when evaluating the effect of each input variable. Fig. 14a indicates that the compressive strength is strongly influenced by water content, the partial dependence decreases almost linearly with the increasing of water content. This is in good agreement with many previous studies on the compressive strength of recycled concrete [41,52,64,112]. Because, in general, adding more water to the concrete mixture, leading to excess free water, and when the concrete becomes hardened, this free water will cause a higher porosity, leading to a reduction in compressive strength. For the case of aggregate content, when aggregate content increases from 0 to around 1000 kg/m^3, the partial dependence decreases almost linearly, this indicates an increase in aggregate content for this rage negatively influences the compressive strength (Fig. 14b). However, when the amount of aggregate rises from 1000 to 2000 kg/m^3, the compressive strength seems to be not affected by aggregate content. Fig. 14c shows the partial dependence of RCA content on the compressive strength. It can be observed that the increase in RCA content from 0 to 1000 kg/m^3 brings a negative impact on the strength, then the increase in RCA content does not affect the compressive strength. This is also consistent with the results found in the previously published works, the increase in RCA content led to a decrease in compressive strength [6,9]. In contrast with RCA content, the increase in the amount of NA content from 0 to 1000 kg/m^3 leads to a significant positive impact on the strength, then when NA content is larger than 1000 kg/m^3, the partial dependence is constant (Fig. 14d). It means that when NA content is larger than 1000 kg/m^3, there is no influence of NA content on the compressive strength. The sand content seems to have little impact on the strength of concrete with RCA (Fig. 14e). The partial dependence increases linearly with cement content, this indicates that cement content affects positively the compressive strength (Fig. 14f). This result of partial dependence is agreed well with the results of feature importance analysis using the SHAP. It can be observed that RCAWA at a lower value has a significant impact on the strength (Fig. 14g), but NAWA does not influence the compressive strength (Fig. 14h). Overall, from Fig. 14, water content and cement content are the most important factors affecting the strength of RCA concrete. In order to quantify the effect range of RCA substitution rate on compressive strength of concrete, the GB_PSO model of the case (iii) is used for PDP 1D. The range of effect is shown in Fig. 14i. The range of substitution rate effect on compressive strength is mean of magnitude value varying from 0 to 5.5. This range is higher than that of RCA content and NA content. Moreover, the variation of substitution rate effect has lower noise than that of RCA and NA content. Therefore, replacing two input variables “RCA content and NA content” with the substitution rate can improve the performance of compressive strength prediction.

In practical engineering applications, the results of PDP 1D can be used for the preliminary mix designs of concrete as the quick calculation of compressive strength can be carried out with the help of Fig. 14. In fact, each graph of Fig. 14 can show a strength gain value due to one component when using the initial mix component. For example, Table 9 shows the compressive strength values prediction of concrete made with different values of RCA substitution rate. Using PDP 1D, the estimation of compressive strength has not obviously reached the high performance of the Gradient Boosting model, however, these estimations seem to be useful in the preliminary study of mix design. (Table 10).

Indeed, among input parameters, there exists a mutual correlation between them, and this mutual also greatly influences the compressive strength of concrete using RCA. Thus, the partial dependence plot (PDP-2D) is implemented to investigate the effect of coupled variables on the strength as well as the interaction among the input variables (Fig. 15). Fig. 15a shows the partial dependence of coupled variables of cement

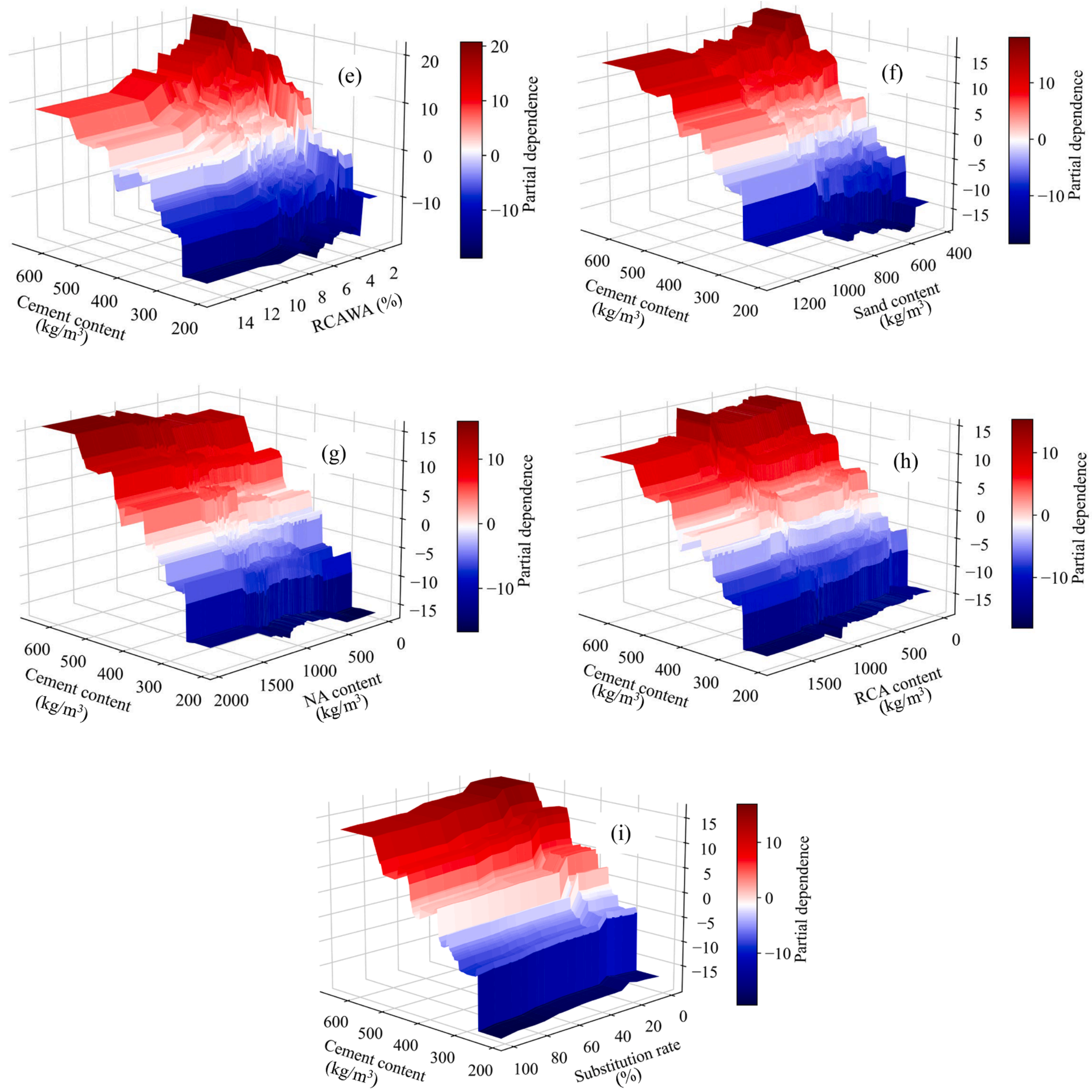


Fig. 15. (*continued*).

and water content. The compressive strength strongly depended on the water content and cement content, the higher cement content and lower water content lead to greater compressive strength. In addition, the results indicate that the dependence of the compressive strength on the cement content was stronger at a lower water content. This is because lower water content and cement content will result in a denser structure of concrete as reported in many previous studies [6,11,41,52]. However, when the water content is lower than 100 (kg/m^3) the dependence of compressive strength on the cement content with cement content higher than 500 (kg/m^3) is not significant. This is because, with very lower water content, the cement hydration process will be limited due to lack of water, as stated in the textbook the water/cement ratio should not be smaller than 0.35 [113]. The dependence of compressive strength on RCAWA was more dominant with a lower water content (Fig. 15b). This is because, high values of RCAWA and RCA created a relatively loose interface between RCA and cement paste in hardened concrete, which results in a lower compressive strength [114]. Fig. 15c shows that the dependence of compressive strength on cement content is not significantly dependent on NAWA content. The dependence of compressive strength on water content is a little strong at a larger NAWA (Fig. 15d). From Fig. 15e, it can be observed that the compressive strength is moderately dependent on the cement content at lower RCAWA. This also can be explained by the smaller porosity of hardened concrete with a smaller RCAWA. The sand content does not affect significantly the dependence of compressive strength on the cement content (Fig. 15f). With a higher NA content, the cement content has a stronger influence on the compressive strength (Fig. 15g). This is because natural aggregate (NA) is known as the load-bearing skeleton of concrete. In contrast, the dependence of compressive strength on cement content is greater at a lower RCA content, this is in alignment with many previous studies [6,7,9,112,114], which reported that the higher amount of RCA, the lower compressive strength (Fig. 15h). The substitution rate has a significant effect in a range from 0 to 20%, out of that range, the substation rate has a small impact on the compressive strength (Fig. 15i). In summary, the results of PDP-2D indicate that cement content and water content are the two most important parameters that affect the compressive strength of concrete with RCA.

5. Conclusions

This study estimated the compressive strength of recycled concrete using six machine learning approaches, including GB, XGB, SVR, GB_PSO, XGB_PSO, and SVR_PSO. The input parameters used for modeling are cement content, water content, aggregate content, natural aggregate content, recycle concrete aggregate content, sand content, water absorption of natural aggregate and RCA.

The result indicates that the performance of hybrid models of GB_PSO, XGB_PSO, and SVR_PSO has higher prediction accuracy than the single models of GB, XGB, and SVR. The GB_PSO has the highest prediction accuracy (R = 0.9356, RMSE = 5.5604 MPa, and MAE = 4.2882 MPa) among three hybrid models.

The best performance of the machine learning model using Gradient Boosting with tuned hyperparameters is achieved and that is one of the main results of this study. Based on the Gradient Boosting with the tuned hyperparameters, soft computing can be developed in predicting the compressive strength of concrete made with RCA.

The results of feature importance analysis using SHAP and partial dependence plot (PDP) analysis revealed that the most important variable effect on compressive strength of concrete made with RAC is cement content, whatever performance strategies of concrete made with RAC.

Based on the partial dependence plot analysis PDP, the content of each material can be computed easily for the required compressive strength. The results also show that the substitution rate of RCA has a higher effect on the compressive strength of concrete than that of RCA and NA content.

In the end, this study provides a systematic evaluation of the compressive strength prediction of recycled concrete and could contribute to the literature and practice of the study on the compressive strength prediction of recycled concrete.

This study mainly focused on main parameters for modeling related to cement content, water content, and aggregate content (natural and recycled aggregates), and water absorption of natural and recycled aggregate, future research should consider more input parameters such as type and content of additive, curing condition, and curing period, etc. Moreover, it is worth noting that the performance of the machine learning model may be enhanced by including more detailed and appropriate variables such as the substitution rate of RAC.

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7. Availability of data and material

Data will be made available on request.

CRediT authorship contribution statement

Van Quan Tran: Data curation, Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing, Validation. **Viet Quoc Dang:** Methodology, Writing – original draft. **Lanh Si Ho:** Conceptualization, Methodology, Visualization, Writing – original draft, Writing – review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.conbuildmat.2022.126578.

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